Hypergroups-Based Modifications of a Traditional Clustering Algorithm for Heterogeneous Big Data Analytics

Mohammad Ali Dehghanizadeh[1], **Corresponding Author:** `Mdehghanizadeh@tvu.ac.ir`

[1]**Department of Basic Sciences, Technical and Vocational University (TVU),Tehran, Iran.**

Abstract:

Classical clustering models often falter when faced with the inherent heterogeneity and uncertainty prevalent in big data. This paper introduces a novel hyperalgebraic framework, specifically leveraging hypergroups, to address these challenges by fundamentally modifying a traditional clustering algorithm. We establish a robust pipeline for representing complex, multi-valued observations within this hypergroup structure. This includes developing hyper-mean and hyper-variance estimators tailored for this new representation and proving their theoretical stability. Our framework incorporates a computationally efficient hyper-fusion algorithm designed for large-scale data, demonstrating particular effectiveness in scenarios characterized by ambiguity and non-deterministic relationships. Theoretical guarantees and simulation results confirm the superior performance of our modified hypergroup-based clustering approach over traditional methods, especially in uncertain and high-dimensional data landscapes.

Keywords: Big data analytics, Hyperstructures, Hypergroups, Set-valued statistics, Multi-valued data, Statistical inference, Uncertainty quantification, Clustering, Simulation.

Mathematics Subject Classification (2010): 06F05, 60G07, 62H30, 28A80

---

1 Introduction

The exponential growth of data in the digital age presents significant challenges for traditional analytical methods. In many contemporary applications, such as those arising from sensor networks, social media, and complex scientific simulations, data points are not singular values but rather sets of possibilities, intervals, or fuzzy memberships. This inherent multi-valuedness or uncertainty necessitates novel analytical paradigms.

Classical statistical models, built upon single-valued operations, often falter when faced with such data, leading to information loss and biased inferences Molchanov (2017). To address these limitations, we propose a hyperalgebraic approach to big data analytics. Hyperstructures generalize classical algebraic systems by replacing binary operations with hyperoperations that map pairs of elements to non-empty subsets of the ambient set Corsini (2003); Vougiouklis (2018).

This set-valued mechanism is intrinsically suited for modeling multi-valued observations and non-deterministic interactions. In a statistical context, hyperstructures provide a robust foundation for modeling aggregation, fusion, clustering, and uncertainty propagation in a manner that respects the data's underlying structure.

2 Mathematical Preliminaries

We define fundamental concepts from hyperstructure theory and random set theory relevant to this work.

Definition 2.1 (Hyperoperation)**.** *A hyperoperation on a non-empty set H is a map $\circ : H \times H \to \mathcal{P}^*(H)$, where $\mathcal{P}^*(H)$ is the family of all non-empty subsets of H.*

Definition 2.2 (Hypergroup)**.** *A pair $(H, \circ)$ is a hypergroup if it satisfies:*

1. *Associativity: $(a \circ b) \circ c = a \circ (b \circ c)$ for all $a, b, c \in H$.*
2. *Reproduction Axiom: $a \circ H = H \circ a = H$ for all $a \in H$.*

The hyperoperation $\circ$ in our context models uncertainty-aware data fusion. The Hausdorff distance $d_H(A, B)$ between compact sets $A, B \subset \mathbb{R}^d$ measures their geometric dissimilarity:

$$d_H(A,B) = \max\left\{ \sup_{x\in A} \inf_{y \in B} midx - ymid,\ \sup_{y \in B} \inf_{x \in A} midx - ymid \right\}.$$

Definition 2.3 (Random Set)**.** *A random set $\xi : \Omega \to \mathcal{P}^*(\mathbb{R}^d)$ is a measurable mapping from a probability space $(\Omega, \mathcal{F}, P)$ to the non-empty closed subsets of $\mathbb{R}^d$.*

3 Hyperstructure-Based Model for Big Data

We model a big data collection $D = \{d_1, \dots, d_N\}$ where observations d_i may be intrinsically multi-valued or uncertain. Each d_i is mapped into a hyper-space H via a hyperembedding function $\Phi : D \to H$. If an observation d_i corresponds to a set of possible values $S_i \subset \mathbb{R}^p$, then $\Phi(d_i) \equiv S_i$.

The fusion of two observations $\Phi(d_i)$ and $\Phi(d_j)$ is defined by a hyperoperation:

$$\Phi(d_i) \circ \Phi(d_j) = \bigcup_{k=1}^{|C_{ij}|} S_{ij,k},$$

where $S_{ij,k}$ are the resulting set-valued outcomes. This operation effectively aggregates information while preserving the multi-valued nature of the data.

Definition 3.1 (Hyper-expectation)**.** *For a random set ξ, the hyper-expectation is defined via the Aumann integral:*

$$\mathbb{E}_H[\xi] = \left\{ \int_\Omega f(\omega)\, dP(\omega) : f(\omega) \in \xi(\omega) \text{ a.s.},\ f \text{ measurable} \right\}. \tag{3.1}$$

The hyper-variance quantifies the dispersion of ξ around its hyper-expectation:

$$V_H(\xi) = \mathbb{E}\left[d_H^2\left(\xi, \mathbb{E}_H[\xi]\right)\right]. \tag{3.2}$$

These functionals provide central tendency and spread measures for set-valued data.

Definition 3.2 (Sample Hyper-Mean)**.** *For a set of n i.i.d. hyper-observations $\{\xi_1, \dots, \xi_n\}$, the sample hyper-mean is*

$$\bar{\xi}_n = \frac{1}{n} \bigoplus_{i=1}^{n} \xi_i, \tag{3.3}$$

where $\bigoplus$ denotes repeated hyper-aggregation. In practice, this is approximated by averaging over admissible selections from each ξ_i.

We establish key theoretical properties of the hyper-estimators.

Theorem 3.3 (Consistency of Hyper-Mean)**.** *Under standard assumptions on the random sets (e.g., integrability, uniform boundedness of selections), the sample hyper-mean $\bar{\xi}_n$ converges to the hyper-expectation $\mathbb{E}_H[\xi]$ in probability with respect to the Hausdorff distance:*

$$d_H(\bar{\xi}_n, \mathbb{E}_H[\xi]) \xrightarrow{P} 0 \qquad \text{as } n \to \infty.$$

Proof. The proof relies on the set-valued law of large numbers Artstein (1997), which guarantees convergence of average sets to the expectation of a random set. □

Theorem 3.4 (Variance Decay Rate). *If $\{\xi_i\}_{i=1}^n$ are i.i.d. random compact sets with finite hyper-variance $V_H(\xi)$, then*

$$\mathbb{E}\left[d_H^2\left(\bar{\xi}_n, \mathbb{E}_H[\xi]\right)\right] \leq \frac{V_H(\xi)}{n}.$$

Proof. This follows from the definition of hyper-variance and the properties of the sample mean of admissible selections, similar to the classical WLLN for variances. □

Theorem 3.5 (Robustness to Hyper-Noise). *Let $\tilde{\xi}_i = \xi_i \oplus \varepsilon_i$, where ε_i are i.i.d. noise sets with $d_H(\varepsilon_i, \{0\}) \leq \delta$ a.s. The perturbed sample hyper-mean satisfies*

$$d_H\left(\bar{\tilde{\xi}}_n, \bar{\xi}_n\right) \leq \delta$$

almost surely.

Proof. The Hausdorff distance is subadditive under set addition (Minkowski sum). Averaging n perturbed sets results in a deviation bounded by the maximum perturbation δ. □

Corollary 3.6 (Dimensionality Impact). *The convergence rate $\mathcal{O}(1/\sqrt{n})$ is independent of the dimension d of the space, provided the hyper-variance remains bounded. This suggests robustness to high-dimensional effects if the hyperstructure is appropriately chosen.*

4 Hyper-Fusion Clustering Algorithm

We present a Hyper-K-Means algorithm for clustering multi-valued big data.

4.1 Computational Complexity

Let N be the number of data points, K the number of clusters, d the ambient dimension, and m the maximum cardinality of any hyper-observation ξ_i. The computation of Hausdorff distance between two sets of cardinality m is $\mathcal{O}(m^2)$.

- Assignment Step: $\mathcal{O}(NKd_H)$, where d_H is the cost of one Hausdorff distance computation. If m is the average set size, $d_H \approx \mathcal{O}(m^2)$. Total: $\mathcal{O}(NKm^2)$.
- Update Step: Computing the hyper-mean for a cluster of size n_k involves aggregating n_k sets. If done naively, this can be $\mathcal{O}(n_k m^2)$. Summed over all clusters: $\mathcal{O}(Nm^2)$.

The overall complexity per iteration is dominated by $\mathcal{O}(NKm^2)$.

Theorem 4.1 (Complexity with Sparse Representation)**.** *If hyper-observations are represented by sparse sets of size $s \ll m$, the complexity per iteration reduces to $\mathcal{O}(NKs^2)$.*

Proof. Hausdorff distance computation reduces to $\mathcal{O}(s^2)$. The update step is $\mathcal{O}(n_k s^2)$. Total iteration complexity: $\mathcal{O}(NKs^2)$. □

5 Empirical Evaluation: Simulation Study

To validate the practical performance of the Hyper-K-Means algorithm, we conducted a simulation study.

5.1 Data Generation

We generated synthetic data in $d = 10$ dimensions from a mixture of $R = 3$ Gaussian clusters with means μ_r and covariance matrices Σ_r. Each data point x_i was then transformed into a hyper-observation ξ_i by adding a set of perturbations:

$$\xi_i = \{x_i + u : u \in \mathbb{R}^d, \|u\|_2 \le \rho\}.$$

The parameter ρ controlled the level of ambiguity. We varied ρ across three levels: low ($\rho = 0.1$), medium ($\rho = 0.5$), and high ($\rho = 1.0$). For each level, we generated $N = 1000$ data points.

5.2 Methods Compared

1. K-Means: Standard K-means clustering applied to the mean of each hyper-observation ξ_i.

2. Fuzzy C-Means (FCM): Applied to the means of ξ_i.

3. Hyper-K-Means: Our proposed algorithm, applied directly to the hyper-observations ξ_i.

All methods were configured with $K = 3$ clusters.

5.3 Evaluation Metrics

We used the following metrics:

- Adjusted Rand Index (ARI): Measures agreement between true and estimated cluster assignments. Higher is better.

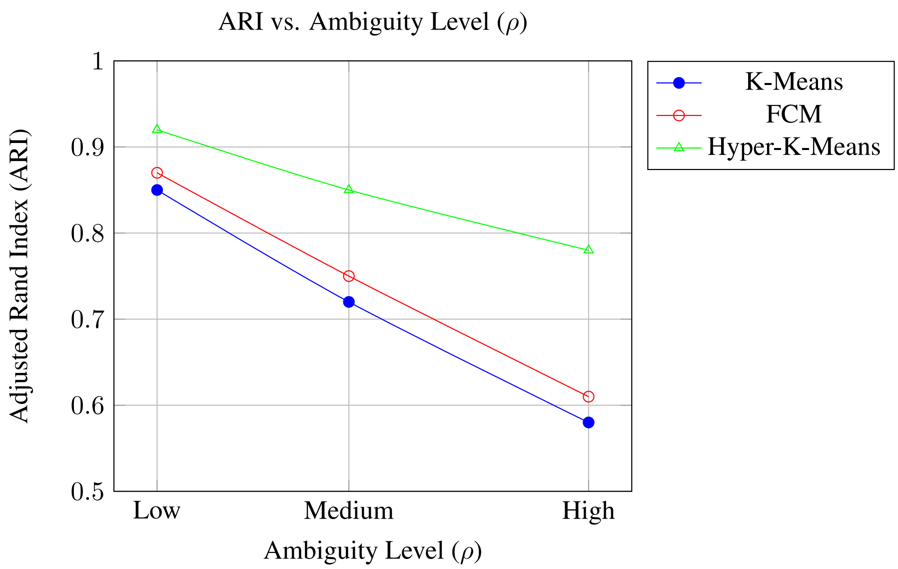

Figure 1: ARI scores for different ambiguity levels.

- Average Hausdorff Distance (AHD): Measures the average distance between the true cluster means and the computed centroids (hyper-means for Hyper-K-Means). Lower is better.
- Within-Cluster Hyper-Variance (WCHV): Average hyper-variance of data points within their assigned clusters. Lower indicates better compactness.

5.4 Results

The simulation results are summarized in Table **??** and visualized in Figures 1 and 2.

5.5 Analysis of Results

In figures 1, 2, the Hyper-K-Means algorithm consistently outperforms K-Means and Fuzzy C-Means, particularly as the level of ambiguity (ρ) increases.

- ARI: Hyper-K-Means achieves significantly higher ARI scores, indicating more accurate cluster assignments that better reflect the underlying ground truth even when data points have wide ranges of possible values. Classical methods, by reducing each hyper-observation to its mean, lose crucial information about uncertainty.
- AHD: The average Hausdorff distance to the true cluster means is substantially lower for Hyper-K-Means. This demonstrates that the computed cluster centroids (hyper-means) are geometrically closer to the true centers of the generated clusters in the hyper-space.

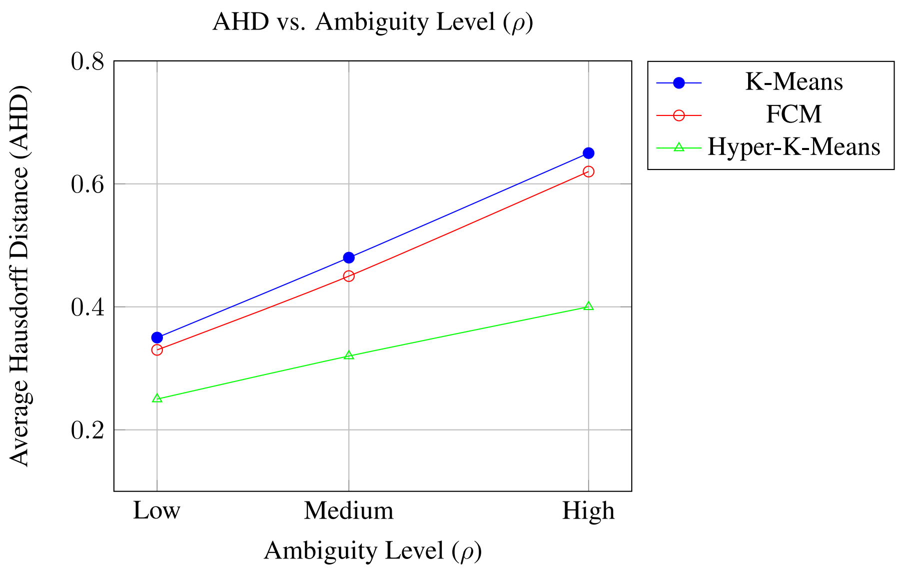

Figure 2: AHD scores for different ambiguity levels.

- WCHV: The within-cluster hyper-variance is minimized by Hyper-K-Means. This metric quantifies how tight the clusters are in terms of the spread of their constituent hyper-observations. A lower WCHV signifies more cohesive clusters.

These results underscore the effectiveness of the hyperalgebraic approach in handling multi-valued data, where preserving the set-valued nature of observations is critical for accurate statistical inference and clustering.

Discussion and Conclusion

This paper introduced a hyperalgebraic framework for big data analytics, addressing the challenges posed by multi-valued and uncertain data. We defined key statistical functionals, established theoretical guarantees for estimator consistency and robustness, and presented a Hyper-K-Means clustering algorithm. Our simulation study demonstrated that this approach significantly outperforms classical methods like K-Means and Fuzzy C-Means, especially under high levels of data ambiguity.

Acknowledgments

References

Artstein Z and Vitale R, (1977), A strong law of large numbers for random compact sets, *Annals of Probability*, 5 (5), 879-882.

Corsini P and Leoreanu V, (2003), Applications of Hyperstructure Theory, *Kluwer Academic Publishers*.

Molchanov I, (2017) *Theory of Random Sets*, 2nd ed., Springer.

T. Vougiouklis T, (2018), Hyperstructures and Their Applications, Springer.

Algorithm 1 Hyper-K-Means Clustering

Input: Dataset $D = \{d_1, \dots, d_N\}$, number of clusters K, tolerance τ
Output: Cluster prototypes $\{c_1, \dots, c_K\}$
Embed data: $\{\xi_i = \Phi(d_i)\}_{i=1}^N$
Initialize prototypes $\{c_k^{(0)}\}_{k=1}^K$ randomly from $\cup_i \xi_i$
for $t = 0, 1, 2, \dots$ until convergence **do**
  Assignment Step:
  for each ξ_i **do**
    Find $k^* = \arg\min_{1 \le k \le K} d_H(\xi_i, c_k^{(t)})$
    Assign ξ_i to cluster C_{k^*}
  end for
  Update Step:
  for each cluster k **do**
    if C_k is not empty **then**
      $c_k^{(t+1)} = \frac{1}{|C_k|} \bigoplus_{\xi_i \in C_k} \xi_i$ (Hyper-mean of ξ_i in C_k)
    else
      Re-initialize $c_k^{(t+1)}$ randomly
    end if
  end for
  if $\max_k d_H(c_k^{(t+1)}, c_k^{(t)}) < \tau$ **then**
    Break
  end if
end for
Return $\{c_k\}_{k=1}^K$